

Classic Mechanics Problems

Trajectory Optimization & Oscillations

Before attempting the problems, please review the following background material:

- **Projectile Motion** — Range, trajectory length, area under curve
- **Drag Forces** — Linear drag models and terminal velocity
- **Small Oscillations** — Linearized equations of motion
- **Heisenberg Uncertainty Principle** — Quantum constraints on classical systems

No calculus is required unless explicitly stated. Students are encouraged to rely on:

- Free-body diagrams
- Symmetry arguments
- Thoughtful choice of reference frame
- Energy and angular momentum conservation where applicable

Conceptual Background

This problem set explores several classic mechanics scenarios that involve optimization and small oscillations. The first three problems deal with projectile motion under gravity, investigating different quantities we might wish to maximize: trajectory length, area under the trajectory, and the effects of drag. The fourth problem examines the stability of circular orbits under small perturbations. The final problem considers a seemingly simple system—a balancing pencil—and reveals a surprising connection to quantum mechanics through the Heisenberg uncertainty principle.

Problem 1. *Maximum trajectory length.*

A ball is thrown at speed v from zero height on level ground. Let θ_0 be the angle at which the ball should be thrown so that the length of the trajectory is maximum. Show that θ_0 satisfies

$$\sin \theta_0 \ln(\sec \theta_0 + \tan \theta_0) = 1$$

You can show numerically that $\theta_0 \approx 56.5^\circ$.

Problem 2. *Maximum trajectory area.*

A ball is thrown at speed v from zero height on level ground. At what angle should it be thrown so that the area under the trajectory is maximum?

Problem 3. *Throwing a beach ball.*

A beach ball is thrown upward with initial speed v_0 . Assume that the drag force from the air is $F_d = -m\alpha v$. What is the speed of the ball, v_f , right before it hits the ground? (An implicit equation is sufficient.) Does the ball spend more time or less time in the air than it would if it were thrown in vacuum?

Problem 4. *Small radial perturbation.*

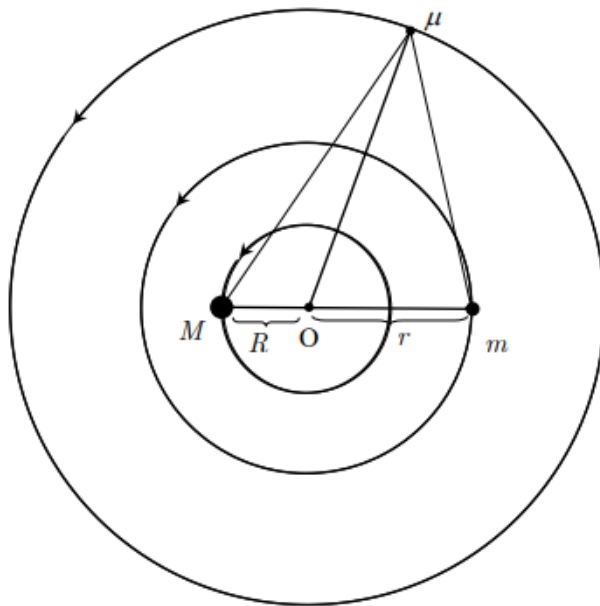


Figure 1: Distance from μ to M is $R + r$ and distance from μ to m is also $R + r$.

Consider the case $M = m$. If μ is now given a small radial perturbation (along $O\mu$), what is the angular frequency of oscillation of μ about the unperturbed position in terms of ω_0 ? Assume that the angular momentum of μ is conserved.

Problem 5. *Balancing a pencil.*

Consider a pencil that stands upright on its tip and then falls over. Let's idealize the pencil as a mass m sitting at the end of a massless rod of length ℓ .

- (a) Assume that the pencil makes an initial (small) angle θ_0 with the vertical, and that its initial angular speed is ω_0 . The angle will eventually become large, but while it is small (so that $\sin \theta \approx \theta$), what is θ as a function of time?
- (b) You might think that it should be possible (theoretically, at least) to make the pencil balance for an arbitrarily long time, by making the initial θ_0 and ω_0 sufficiently small. However, it turns out that due to Heisenberg's uncertainty principle (which puts a constraint on how well we can know the position and momentum of a particle), it is impossible to balance the pencil for more than a certain amount of time. The point is that you can't be sure that the pencil is initially both at the top and at rest. The goal of this problem is to be quantitative about this. The time limit is sure to surprise you.

Without getting into quantum mechanics, let's just say that the uncertainty principle says (up to factors of order 1) that

$$\Delta x \Delta p \gtrsim \hbar, \quad \text{where} \quad \hbar \approx 1.05 \times 10^{-34} \text{ J} \cdot \text{s}$$

is Planck's constant. The implications of this are somewhat vague, but we'll just take it to mean that the initial conditions satisfy

$$m \ell \omega_0 \cdot \ell \theta_0 \gtrsim \hbar.$$

With this constraint, your task is to find the maximum time it can take your $\theta(t)$ solution in part (a) to become of order 1. In other words, determine (roughly) the maximum time the pencil can balance. Assume $m = 0.01$ kg, and $\ell = 0.1$ m.